

Full Stack Development with ServiceNow

Project Documentation Format

Field	Details
Project Title	Software Installation Request Automation in ServiceNow
Project Type	Individual Project
Technology	ServiceNow Platform
Core Areas	Service Catalog, Workflow/Flow Automation, Business Rules, Catalog Tasks, Notifications, Update Sets
Objective	Standardize and automate software installation requests from submission through approval, fulfillment, tracking and license-related exception handling.

1. Introduction

- Project Title: Software Installation Request Automation in ServiceNow
- Project Type: Individual Project
- Developer: Talamala Mercy
- Platform: ServiceNow

This project automates the software installation request process using ServiceNow. Instead of relying on informal email or verbal requests, an employee submits a standardized Service Catalog request. The request is processed through configured workflow/automation, approval and fulfillment activities, with request/task tracking and a defined response to license shortages.

2. Project Overview

Purpose: The purpose of the project is to provide a structured and transparent process for requesting software installation. The solution reduces manual coordination, improves visibility, and enables IT teams to process requests consistently.

Features:

- Software Installation Request Service Catalog item.
- Structured variables for Software Name, Version Required, License Justification and Urgency.
- Workflow/automation for approval and request processing.
- Automated catalog task creation for the Software Support / Fulfillment Team.
- Request, requested-item and catalog-task tracking using ServiceNow tables.
- Notifications for requester, approver and IT staff.
- SLA-oriented fulfillment tracking.
- Business Rule to create an incident when a license shortage causes an On Hold condition.
- Update Set export/import for configuration transfer between instances.

3. Architecture

Frontend / User Experience: The user-facing layer is the ServiceNow Service Catalog and, where used, the Service Portal experience. Employees interact with a catalog item and submit structured request data.

Process / Automation: ServiceNow Workflow/Flow Automation handles approval and fulfillment steps, while Notifications communicate lifecycle updates. Business Rules handle specific server-side logic such as license-shortage exception processing.

Data: The solution uses ServiceNow tables including sc_request, sc_req_item and sc_task for the request lifecycle. Incident records are used for license-related exceptions.

Layer	ServiceNow Component	Purpose
User Experience	Service Catalog / Service Portal	Submit and view software installation requests
Automation	Workflow / Flow	Approval, task generation and lifecycle automation
Server-side Logic	Business Rule	License-shortage exception handling
Request Data	sc_request	Stores the parent request
Requested Item	sc_req_item	Stores the catalog item/request details
Fulfillment	sc_task	Stores IT fulfillment work
Issue Management	Incident	Tracks license-shortage exception
Deployment	Update Set	Packages and transfers configuration

4. Setup Instructions

Prerequisites:

- Access to a ServiceNow development/test instance.
- Required administrative/developer permissions to configure Service Catalog and automation.
- The Software Installation Request project configuration or a clean instance for implementation.

Installation / Configuration:

1. Open ServiceNow and create a local Update Set named for the Software Installation Request project.
2. Create the Software Installation Request catalog item.
3. Create the catalog variables: Software Name, Version Required, License Justification and Urgency.
4. Test the catalog item using Try It and/or the Service Portal.
5. Create the configured Workflow/Flow automation with the required approval and task activities.
6. Attach the workflow/flow to the catalog item.
7. Verify sc_request, sc_req_item and sc_task records after submission.
8. Create the Business Rule used for license-shortage handling.
9. Complete functional testing and move the Update Set to Complete.
10. Export the Update Set to XML for deployment to another instance.

5. Application Structure

Because this is a ServiceNow application rather than a MERN application, the project is organized around ServiceNow configuration objects instead of client/server folders.

Area	Configuration Objects
Data	Catalog item, catalog variables, sc_request, sc_req_item, sc_task
Automation	Workflow / Flow, approval activity, task activity
Business Logic	Business Rule for license-shortage exception
User Experience	Service Catalog and Service Portal request experience
Communication	ServiceNow notifications
Reporting / Monitoring	Request, task and SLA tracking views
Deployment	Update Set

6. Running the Application

The solution runs inside the ServiceNow instance. No local npm start command is required.

- Log in to the ServiceNow instance.
- Open the Service Catalog.
- Search for the Software Installation Request catalog item.
- Open the item, enter the required variables, and submit the request.
- Open the generated request to review the request and requested-item records.
- Verify the generated catalog task and its assignment/state.
- Observe the configured approval, automation and notifications.
- For license-shortage testing, place the requested item On Hold and verify the Business Rule creates the incident.

7. Automation / Interface Documentation

The ServiceNow project does not expose a custom Node/Express REST API in the supplied project document. The primary interfaces are ServiceNow records, catalog submissions and automation actions.

Interface / Record	Method / Action	Input	Expected Outcome
Service Catalog Item	Submit / Order Now	Software Name, Version Required, License Justification, Urgency	Request is created
sc_request	Create / View	Requested user and request details	Parent request record is available
sc_req_item	Create / View	Catalog item variables	Requested item contains submitted values
sc_task	Create / Update	Assignment, short description, state	Fulfillment task is generated and tracked
Workflow / Flow	Execute	Request / requested item context	Approval and fulfillment automation runs
Business Rule	Trigger	On Hold / license-shortage condition	Incident is created
Update Set	Export / Import / Commit	Configuration changes	Configuration can be transferred

8. Authentication

Authentication is handled by the ServiceNow instance and its configured user access controls. Users must sign in to the ServiceNow environment before accessing the catalog and project records.

Authorization is controlled through ServiceNow roles, groups and access permissions. The project uses the Software Support / Fulfillment Team for catalog task handling, while approval responsibilities are handled through the configured approval process.

No custom JWT or session-token mechanism is described in the supplied project document.

9. User Interface

The main user interface is the Service Catalog request form. The project source specifies the catalog variables Software Name, Version Required, License Justification and Urgency.

UI Screen	Main Elements
Software Installation Request	Catalog item name, request description and variables
Request Form	Software Name, Version Required, License Justification, Urgency
Service Portal	Search and submit the same catalog item
Request Record	Request number and request lifecycle
Requested Item	Catalog variable values and item status
Catalog Task	Assignment to Software Support, short description and task state

10. Testing

Testing covers the functional workflow and repeated operational execution of the project.

Test ID	Test Area	Expected Result
FT-01	Catalog Item	Software Installation Request item opens correctly
FT-02	Catalog Variables	All required variables accept valid input
FT-03	Request Creation	sc_request and sc_req_item records are created
FT-04	Automation	Configured approval/workflow executes
FT-05	Catalog Task	Task is created, assigned and trackable
FT-06	Task State	Open / Work in Progress / Completed states are handled as configured
FT-07	License Shortage	On Hold condition triggers the intended incident creation
FT-08	Update Set	Configuration can be exported and transferred

Performance consideration: the same test sequence should be repeatable across representative requests without record loss, task loss or automation failure.

11. Screenshots or Demo

Capture the following ServiceNow screens for project demonstration:

- Software Installation Request catalog item.
- Catalog variables and sample request data.
- Service Portal search and request submission.
- Generated Request / REQ number.
- Requested Item record showing populated variables.
- Catalog Task assigned to Software Support.
- Task state during fulfillment.
- License-shortage Business Rule result and generated incident.
- Completed Update Set, export XML, retrieve, preview and commit.

A live demo should follow the same request-to-fulfillment lifecycle.

12. Known Issues

- The exact approval rules depend on the configured workflow/flow and organizational policy.
- License availability is an external dependency; the project handles the shortage as an exception but cannot create a license.
- The supplied project document refers to Workflow Editor steps; ServiceNow releases/environments may present the equivalent capability through Flow Designer/Workflow interfaces.
- Actual final screenshots, production incident counts and live deployment URLs were not provided in the supplied project document and therefore are not claimed here.

13. Future Enhancements

- Integrate license or asset data so availability can be checked automatically before fulfillment.
- Add approval routing based on department, software type, cost, risk or requester role.
- Add automatic reminders and escalation for delayed approvals or fulfillment tasks.
- Create dashboards for request volume, SLA compliance, fulfillment time and license issues.
- Integrate CMDB/endpoint information to validate the target device.
- Add policy validation for restricted or high-risk software.
- Extend the catalog to additional IT software and access-request processes.

Project Mapping to Supplied ServiceNow Project Document

This FSD documentation is derived from the supplied Software Installation Request Automation project. The project document identifies the objective, Service Catalog item, catalog variables, workflow, catalog tasks, catalog tables, Business Rule, testing and Update Set deployment as the main implementation areas.

Source Project Area	FSD Section
Service Catalog & Variables	2. Project Overview; 4. Setup Instructions; 9. User Interface
Workflow / Approval	3. Architecture; 7. Automation / Interface Documentation; 10. Testing
sc_request / sc_req_item / sc_task	3. Architecture; 7. Automation / Interface Documentation
Business Rule / License Shortage	3. Architecture; 10. Testing; 12. Known Issues
Update Sets	4. Setup Instructions; 7. Automation / Interface Documentation; 11. Screenshots or Demo